

THERMODYNAMICS-BASED OPTIMIZATION AND CONTROL OF VAPOR-COMPRESSION CYCLE OPERATION: CONTROL SYNTHESIS

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ABSTRACT

This paper considers the implementation of an exergy-based multiple degree of freedom (MDOF) optimization and control methodology for the operation of VCC systems. The optimization problem for the standard VCC is characterized in terms of 4 thermodynamic variables and 1 fluid-dynamic variable. The resulting control problem is then analyzed, and a design variable, Λ , is introduced which allows the user to choose how the optimization variables are projected onto a control space of lower dimension. The potential of this approach to improve operational efficiency, with respect to both first and second law efficiency metrics, is demonstrated on an experimental VCC system through implementation of the proposed optimization using a feedforward plus feedback control architecture.

NOMENCLATURE

a	aperture
C	cooling capacity
h	specific enthalpy
K	flow coefficient, gain matrix
$\dot{m}$	mass flow rate
P	pressure
$\dot{Q}$	heat transfer rate
T	temperature
$\dot{W}$	work transfer rate
δ	displacement
ρ	density
ω	rotational speed
X	exergy

Subscript

1,2,3,4	transition points of the VCC (applies to thermodynamic state variables)
a	air
c	condenser
e	evaporator
H	high-temperature reservoir
k	compressor
L	low-temperature reservoir
r	refrigerant
v	electronic expansion valve
vol	volumetric

1 INTRODUCTION

Vapor-compression cycle (VCC) systems are used to remove heat from a low-temperature environment and reject it to a high-temperature reservoir, typically ambient air. They are generally designed to operate optimally at a particular cooling load condition, which is termed the ‘design point’ of the system. However, in practice, VCC systems are often operated away from their design point due to either variations in cooling load or the installation of a system which has the capability of meeting some peak cooling load albeit the nominal one may be significantly less. As a result, on-off control strategies applied to a fixed-speed compressor are typically utilized to operate a VCC system at partial-load conditions which, in turn, compromise operational efficiency.

Electronic actuators, such as variable-speed compressors, electronic expansion valves (EEVs), and variable-speed fans, have long been advocated as integral to improving the efficiency

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of VCC systems [1–3]. Moreover, these actuators provide the control engineer with access to more degrees of freedom (DOFs) and consequently, the ability to operate the system at partial-load conditions without cycling the compressor. As a result there is significant potential for improving the off-design point operational efficiency of any given VCC system through optimization and control of these available DOFs.

Many optimization approaches have been proposed for VCC systems [4–6] in which the objective function to be minimized is empirically-derived and specific to a particular system design. Additionally, the number of DOFs considered is often constrained *a priori* [4, 7, 8]. This motivated the development of a thermodynamics-based multiple degree of freedom (MDOF) optimization framework for optimal steady-state operation of the vapor-compression cycle (VCC) [9]. The DOFs of the VCC were characterized as the three enthalpies in the cycle, h_1 , h_2 , h_4 , one cycle temperature, T_1 , and the refrigerant mass flow rate, $\dot{m}_r$. The objective function was constructed to minimize the rate of exergy destruction [10, 11] in the cycle, a combined first and second-law based approach to efficiency, as opposed to minimizing power consumption alone. A simulated case study was conducted in [9] to compare operation at optimized set points, obtained by minimizing the exergy-based objective function, against operation data collected on a commercial truck transport refrigeration system (TTRS). The optimized operation set points suggested an improvement of 52.5% in the coefficient of performance (COP) of the TTRS. However, the optimization was not verified on an experimental system.

In this paper the authors consider the control design and implementation associated with this exergy-based MDOF optimization framework for steady-state operation of the standard VCC system. The control problem is complicated by the fact that the optimization space is 5-dimensional whereas the controllable space is only 4-dimensional. The proposed optimization and control architecture to address this challenge is motivated by a dynamical understanding of the system. Through implementation on an experimental system, improvements in system efficiency with respect to both the first and second laws of thermodynamics are demonstrated.

The paper is organized as follows. A background of the VCC and the exergy-based objective function is given in Sec. 2. In Sec. 3 the control problem is formulated and analyzed. In Sec. 4, an experimental case study is presented, followed by analysis and discussion in Sec. 5. Finally, conclusions of this work are summarized in Sec. 6.

2 BACKGROUND

2.1 Vapor-Compression Cycle

The standard vapor-compression cycle (VCC) consists of four processes of the refrigerant: compression (1 to 2), condensation (2 to 3), expansion (3 to 4), and evaporation (4 to 1), where 1, 2, 3 and 4 refer to the transition points of the cycle defined in Fig. 1. The standard VCC assumes

- isobaric condensation and evaporation
- isenthalpic expansion
- isentropic compression
- evaporation of refrigerant to a saturated or superheated vapor state
- condensation of refrigerant to a saturated or subcooled liquid state

The coefficient of performance (COP) for the standard VCC op-

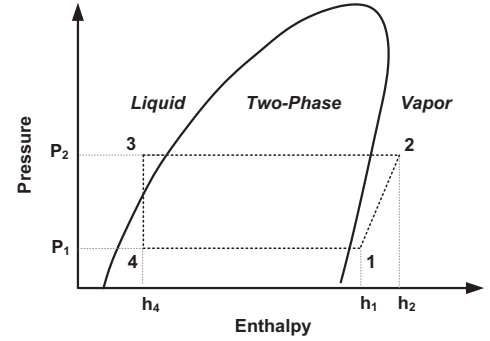


Figure 1. PRESSURE-ENTHALPY (P - h) DIAGRAM FOR A VAPOR-COMPRESSION CYCLE WITH FOUR THERMODYNAMIC DEGREES OF FREEDOM.

erating at steady-state is given in Eq. (1) [12]. The COP is an efficiency metric based on the first law of thermodynamics. A higher COP corresponds to more efficient operation, and its upper bound is a function of the temperatures of the high and low-temperature reservoirs between which the cycle is operating [12].

$$\text{COP} = \frac{\text{evaporator capacity}}{\text{supplied power}} = \frac{\dot{Q}_L}{\dot{W}} = \frac{\dot{m}_r(h_1 - h_4)}{\dot{m}_r(h_2 - h_1)} < \frac{T_L}{T_H - T_L} \quad (1)$$

An additional metric, the second law (exergetic) efficiency, η_{II} [13], provides a more complete measure of the efficiency of the VCC. Intuitively, η_{II} tell us how “close to ideal” the VCC is operating.

$$0 \leq \eta_{II} = \left\{ 1 - \frac{\text{exergy destroyed}}{\text{exergy supplied}} \right\} = \frac{\text{COP}}{\left(\frac{T_L}{T_H - T_L} \right)} < 1 \quad (2)$$

2.2 MDOF Objective Function

Based on the constitutive relationships between pressure, temperature, entropy, etc. [11] and the assumptions listed in Sec. 2.1, the VCC has four thermodynamic DOFs. The three enthalpies of the cycle, $\{h_1, h_2, h_3 = h_4\}$, and any one of the following, $\{P_1, P_2, T_1\}$, uniquely define the remaining thermodynamic states at each of the transition points, denoted 1–4 in Fig. 1.

Moreover, there is an additional fifth DOF: the refrigerant mass flow rate, $\dot{m}_r$. This DOF is a fluid dynamic variable, rather than a thermodynamic one, and is not captured in a P - h diagram of the cycle.

The exergy-based MDOF objective function [9] to be minimized is given in Eqs. (3)–(5).

$$J = (\gamma)J_1 + (1 - \gamma)J_2 \quad (3)$$

$$J_1 = |C_{desired} - C_{achieved}| = |C_{desired} - \dot{m}_r(h_1 - h_4)| \quad (4)$$

$$J_2 = \dot{X}_{destroyed} = \dot{m}_r(h_1 - h_4) \left(1 - \frac{T_H}{T_L}\right) + \dot{m}_r(h_2 - h_1) \quad (5)$$

The performance of the cycle is captured in J_1 , Eq. (4), the absolute value of the error between the desired cooling capacity, $C_{desired}$, and achieved cooling capacity, $C_{achieved}$. The efficiency of the cycle is captured in J_2 , Eq. (5), the rate of exergy destruction in the cycle [10]. Rather than explicitly constraining the achieved cooling capacity, a weighting factor, γ , is introduced to allow the user to weigh the tradeoff between performance and efficiency.

All but one of the five DOFs appears explicitly in the objective function; T_1 instead explicitly appears in the thermodynamic constraints associated with J :

1. $h_1 < h_2$,
2. $h_4 < h_1$,
3. $T_1 \leq T_L$,
4. $T_3 \geq T_H$,
5. $T_1 - T_4 \geq 0$,
6. $T_{3,sat} - T_3 \geq 0$.

Note that $T_{3,sat}$ is the saturated refrigerant temperature at P_2 . The first two constraints ensure that compression and evaporation of the refrigerant, respectively, occur. The third and fourth constraints ensure the correct temperature gradients at the outlet of the evaporator and condenser, respectively. The fifth constraint ensures that only refrigerant vapor is compressed. Finally, the sixth constraint ensures that only refrigerant liquid is expanded.

3 CONTROL ANALYSIS

In practice, the processes of the VCC are realized by a compressor, condenser, expansion device (e.g. electronic expansion valve), and evaporator, as shown in Fig. 2. The control inputs for a standard VCC system are electronic expansion valve (EEV) aperture, a_v , compressor speed, ω_k , and the evaporator and condenser fan speeds, specified here as a percentage of the total power input to each fan, $\dot{W}_{f,e}$ and $\dot{W}_{f,c}$, respectively. It is assumed that the compressor and evaporator and condenser fans are variable-speed. We define the vector of control input variables as

$$\mathbf{u} = (a_v \ \omega_k \ \dot{W}_{f,e} \ \dot{W}_{f,c})^T \in \mathfrak{R}^4. \quad (6)$$

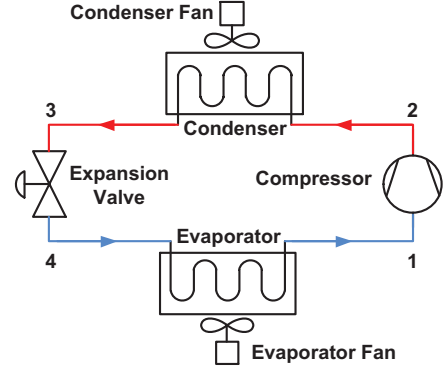


Figure 2. SCHEMATIC OF VAPOR-COMPRESSION CYCLE SYSTEM.

The four thermodynamic optimization variables, h_1 , h_2 , h_4 , and T_1 are rewritten via constitutive relationships [11] in terms of an equivalent set of pressures and temperatures which can be measured on a physical system. Thus we can redefine the complete vector of optimization variables as

$$\mathbf{v} = (P_1 \ P_2 \ T_1 \ T_3 \ \dot{m}_r)^T \in \mathfrak{R}^5 \quad (7)$$

where

$$\mathbf{v}^* = \arg \min_{\mathbf{v}} (J) = (P_1^* \ P_2^* \ T_1^* \ T_3^* \ \dot{m}_r^*)^T \in \mathfrak{R}^5. \quad (8)$$

The steady-state behavior of the standard VCC system is described by the following 9 equations:

$$\hat{K}_v = f_1(u_1, P_1, P_2) \quad (9)$$

$$\hat{\eta}_{vol} = f_2(u_2, P_1, P_2) \quad (10)$$

$$(\widehat{UA})_L = f_3(u_3, P_1, P_2, \dot{m}_r) \quad (11)$$

$$(\widehat{UA})_H = f_4(u_4, P_2, \dot{m}_r) \quad (12)$$

$$\hat{Q}_L = (\widehat{UA})_L (T_L - T_4) = \dot{m}_r(h_1 - h_4) \quad (13)$$

$$\hat{Q}_H = (\widehat{UA})_H (T_3 - T_H) = \dot{m}_r(h_2 - h_4) \quad (14)$$

$$\hat{m}_{r,v} = \hat{K}_v \sqrt{\rho_3(P_2 - P_1)} \quad (15)$$

$$\hat{m}_{r,k} = \hat{\eta}_{vol} \delta_k u_2 \rho_1 \quad (16)$$

$$\hat{m}_{r,v} = \hat{m}_{r,k} = \dot{m}_r \quad (17)$$

where K_v is the valve flow coefficient, η_{vol} is the compressor volumetric efficiency, and $(UA)_L$ and $(UA)_H$ are the overall heat transfer coefficients [14] at the low and high-temperature reservoirs, respectively. The functions f_1 , f_2 , f_3 , and f_4 are empirically-derived nonlinear relationships. The hat notation is used to denote that $\hat{\mathbf{u}}^*$ is obtained based on the functions f_1 – f_4

which have some associated uncertainty; e.g.

$$|K_v - \hat{K}_v| = \tilde{K}_v \geq 0 \quad (18)$$

This uncertainty is further propagated to the calculation of $\dot{Q}_L$, $\dot{Q}_H$, $\dot{m}_{r,v}$, and $\dot{m}_{r,k}$. In the event that the model uncertainty associated with f_1 – f_4 is zero, $\hat{\mathbf{u}}^* = \mathbf{u}^*$, where $\mathbf{u}^*$ is the actual input vector required to achieve operation at the optimal set points, $\mathbf{v}^*$. Therefore, the elements of $\hat{\mathbf{u}}^*$ are analogous to model-based feedforward control input signals (Fig. 3).

When the model uncertainty is nonzero,

$$|\mathbf{u}^* - \hat{\mathbf{u}}^*| = \tilde{\mathbf{u}} \geq 0. \quad (19)$$

One method for compensating for $\tilde{\mathbf{u}}$ is through the use of feedback control. Even with some level of model uncertainty, the feedforward control input signals, $\hat{\mathbf{u}}^*$, which are computed using *nonlinear* static models, move the system close to the operation set-points specified by $\mathbf{v}^*$. This allows for a *linear* feedback controller to locally augment the feedforward control input signals so that operation at $\mathbf{v}^*$ is achieved.

However, a consequence of the fluid dynamics of the EEV and compressor is that the underlying 4-input 5-output problem is uncontrollable. As long as u_1 and u_2 and any two of the following three variables — P_1 , P_2 , $\dot{m}_r$ — are specified, the third variable will be constrained by Eqs. (15)–(17). Instead, one must project $\mathbf{v} \in \mathfrak{R}^5$ from the 5-dimensional optimization space onto the 4-dimensional control space, resulting in a new vector $\mathbf{y} \in \mathfrak{R}^4$ that is some linear combination of the original optimization variables (Eq. (20)). The matrix Λ is a design variable that can be chosen based on a variety of factors, including sensor cost, sensor accuracy, or sensitivities in the nonlinear empirically-derived functions f_1 – f_4 . A schematic of the complete optimization and control architecture is shown in Fig. 3.

$$\mathbf{y}^* = \Lambda \mathbf{v}^* \quad (20)$$

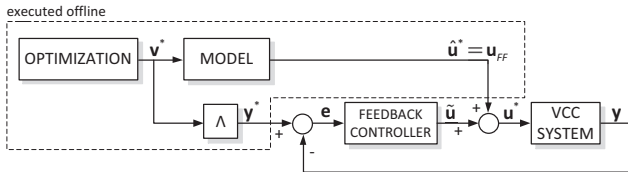


Figure 3. SCHEMATIC OF OPTIMIZATION AND CONTROL ARCHITECTURE.

4 EXPERIMENTAL CASE STUDY

To validate the improvement in efficiency achievable with this approach, we will present a case study in which the objective is to track two different steady-state cooling capacities, as shown in Fig. 4, on an experimental system. The load profile begins with a desired capacity of 0.70kW (Case 1) and then increases to 0.85kW (Case 2).

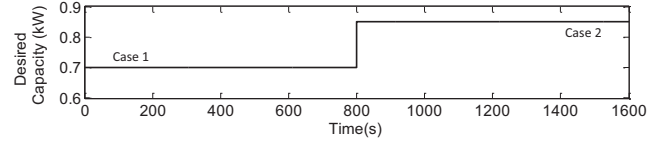


Figure 4. COOLING CAPACITY LOAD PROFILE.

4.1 System Description

The experimental system is a VCC training stand at the University of Illinois at Urbana-Champaign. The system is configured with a semi-hermetic 0.6 hp reciprocating compressor, a condenser with receiver, an EEV, and an evaporator, as well as condenser and evaporator fans, and it operates with refrigerant R134a. A complete description of the system is available in [15].

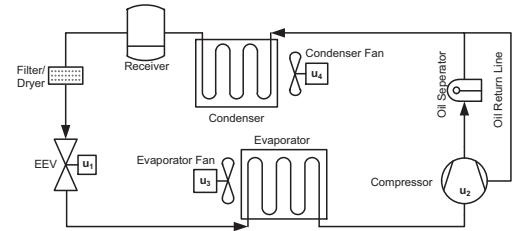


Figure 5. SCHEMATIC OF EXPERIMENTAL SYSTEM.

4.2 Offline Optimization

4.2.1 Constraints The experimental system contains a receiver tank at the outlet of the condenser as shown in Fig. 5, thereby constraining the condenser outlet refrigerant condition to be saturated liquid:

$$T_3 = T_{3,sat}. \quad (21)$$

Furthermore, for clarity of exposition, we assume that the heat transfer across each heat exchanger is maximized. This is equiv-

alent to imposing the following constraints:

$$\dot{Q}_{L,max} = (UA)_{L,max}(T_L - T_4) = \dot{m}_r(h_1 - h_4) \quad (22)$$

$$\dot{Q}_{H,max} = (UA)_{H,max}(T_3 - T_H) = \dot{m}_r(h_2 - h_4). \quad (23)$$

The maximum overall heat transfer coefficient is estimated using heat transfer correlations and a thermal resistance circuit [9].

From a control perspective, these constraints are satisfied by operating the evaporator and condenser fans, respectively, at their maximum power such that the overall heat transfer coefficients of the evaporator and condenser, $(UA)_L$ and $(UA)_H$, are maximized. Therefore, the control space is constrained to be 2-dimensional: $\mathbf{u} = (a_v, \omega_k)^T$. The final optimization problem is defined by 5 DOFs — $\mathbf{v} = (P_1, P_2, T_1, T_3, \dot{m}_r)^T$ — with three constraints given by Eqs. (21), (22), and (23).

4.2.2 Solution The inputs to the optimization, in addition to the desired cooling capacity, are

1. $T_L = 11.6^\circ\text{C}$
2. $T_H = 23.2^\circ\text{C}$
3. $\gamma = 0.4$.

The weighting factor, γ , was chosen heuristically to sufficiently weigh J_1 so as to achieve the desired capacity. The upper and lower bound constraints on the optimization variables are given in Table 1 and are based on the fluid properties of R134a [16] as well as the design of the experimental system components [17].

Table 1. UPPER AND LOWER BOUND CONSTRAINTS

		Lower Bound	Upper Bound
P_1	(kPa)	100	1200
P_2	(kPa)	100	1200
T_1	($^\circ\text{C}$)	-20	50
$\dot{m}_r$	(kg/s)	0.00	0.0065

The optimal solutions for the two cases, Case 1: $C_{desired} = 0.7\text{kW}$ and Case 2: $C_{desired} = 0.85\text{kW}$, obtained using the function *fmincon* in the MATLAB Optimization Toolbox [18], are given in Tab. 2.

The feedforward control inputs for $u_1 = a_v$ and $u_2 = \omega_k$ are determined by iteratively solving Eqs. (15)–(17); the empirical models for $\hat{K}_v$ and $\hat{\eta}_{vol}$ for the particular EEV and compressor on the experimental system are given in [17]. As noted above, the feedforward control inputs for $u_3 = \dot{W}_{f,e}$ and $u_4 = \dot{W}_{f,c}$ are constrained to be their maximum possible value (Tab. 3).

Table 2. OPTIMIZATION RESULTS

	Units	Case 1	Case 2
Capacity Achieved	kW	0.700	0.850
Optimal Cost, J	kW	0.0483	0.0728
P_1	kPa	261	235
P_2	kPa	818	867
T_1	$^\circ\text{C}$	2.79	-0.0123
T_3	$^\circ\text{C}$	32.1	34.2
$\dot{m}_r$	kg/s	0.00447	0.00560

Table 3. MODEL-BASED FEEDFORWARD CONTROL INPUTS

		Desired Capacity	
	Units	0.70kW	0.85kW
a_v	%	9.5	11
ω_k	rpm	950	1300
$\dot{W}_{f,e}$	%	100	100
$\dot{W}_{f,c}$	%	100	100

4.3 Feedback Control Design

By virtue of the constraints described by Eqs. (21)–(23), only two variables in $\mathbf{v}$ can be independently controlled. Therefore, $\mathbf{v} \in \mathfrak{R}^5$ will be projected onto $\mathbf{y} \in \mathfrak{R}^2$. The particular choice of Λ , Eq. (24), was based on the superior time response and accuracy of pressure transducers as compared to that of thermocouples and mass flow sensors on the experimental system considered in this case study.

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \\ T_1 \\ T_3 \\ \dot{m}_r \end{pmatrix} \quad (24)$$

A feedback controller is used to compensate for $\tilde{u}_1 = u_1^* - u_{1,FF}$ and $\tilde{u}_2 = u_2^* - u_{2,FF}$ to regulate $\mathbf{y}$. The desire to track references with minimal RMS error leads to the choice of a 2-norm based optimal controller such as a linear quadratic regulator (LQR). However, this is just one of many possible feedback controllers that could be used.

A standard prediction error/maximum likelihood system identification (ID) algorithm in the MATLAB System ID Toolbox [19] was used to identify a second-order, 2-input 2-output linear state space mapping between $\mathbf{u}$ and $\mathbf{y}$. Details of the identification are included in Appendix A. Since the identified system order is the same as the number of inputs and outputs, the origi-

nal state space realization of the plant, Eq. (38), was transformed by replacing x with $\tilde{C}^{-1}\bar{y}$, as shown in Eqs. (25)-(27), such that the output vector, $\bar{y}$, is treated as the new state vector of the system. Note that $\bar{y}$ and $\bar{u}$ represent scaled vectors; x was not scaled during the system ID process. Moreover, all variables represent deviation variables from the nominal operating condition given in Appendix A.

$$\dot{\bar{y}} = \tilde{A}\bar{y} + \tilde{B}\bar{u} \quad (25)$$

$$\bar{y} = \tilde{C}\bar{y} \quad (26)$$

$$\tilde{A} = \tilde{C}\tilde{A}\tilde{C}^{-1}, \tilde{B} = \tilde{C}\tilde{B}, \tilde{C} = I_2 \quad (27)$$

The original system is Type 0 and will therefore only track step reference inputs with a non-zero static error constant [20]. Consequently, the scaled identified plant model was augmented with two integrators to ensure zero steady-state error when the system is required to track step references, as shown in Eq. (28), where $\bar{y} \in \mathbb{R}^2$ represents the new plant states, $\bar{z} \in \mathbb{R}^2$ represents the additional integrator states, and $\bar{u} \in \mathbb{R}^2$ represents the input vector.

$$\begin{bmatrix} \dot{\bar{y}} \\ \dot{\bar{z}} \end{bmatrix} = \begin{bmatrix} \tilde{A} & 0 \\ \tilde{C} & 0 \end{bmatrix} \begin{bmatrix} \bar{y} \\ \bar{z} \end{bmatrix} + \begin{bmatrix} \tilde{B} \\ 0 \end{bmatrix} \bar{u} \quad (28)$$

The linear quadratic cost function, J_{LQR} , is shown in Eq. (29). The weighting matrices, Q_P and Q_I , were tuned to improve the transient performance of the controller; R was tuned to ensure that the actuation did not cause excessive wear on the valve and compressor (Eq. (30)). The resulting gain matrix, K , is given in Eq. (31). Note that $K \in \mathbb{R}^{2 \times 4}$ as a result of 2 control inputs and 4 states (2 plant states and 2 additional integrator states).

$$J_{LQR} = \int_0^\infty \left((\bar{y} - \bar{y}^*)^T Q_P (\bar{y} - \bar{y}^*) + (\bar{z} - \bar{z}^*)^T Q_I (\bar{z} - \bar{z}^*) + \bar{u}^T R \bar{u} \right) dt \quad (29)$$

$$Q_P = \begin{bmatrix} 50 & 0 \\ 0 & 50 \end{bmatrix}, Q_I = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, R = \begin{bmatrix} 250 & 0 \\ 0 & 200 \end{bmatrix} \quad (30)$$

$$K = \begin{bmatrix} 5.08 & -1.07 & 0.0833 & -0.0230 \\ 3.91 & 3.22 & 0.0364 & 0.0659 \end{bmatrix} \quad (31)$$

4.4 Experimental Results

The optimization and control architecture shown in Fig. 3 was implemented on the experimental system described in Sec. 4.1. Figure 6 shows the regulation of the optimized set points; note that only linear combinations of P_1 and P_2 were regulated with the feedback controller based on the choice of Λ , Eq. (24). Figure 7 shows the control inputs of the feedback plus feedforward control architecture as compared to feedforward alone.

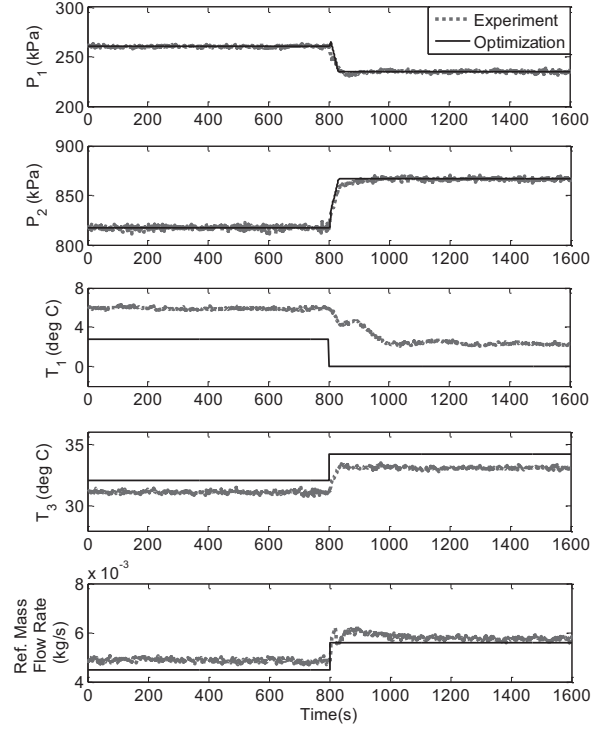


Figure 6. OPTIMIZATION VARIABLES PLOTTED AS A FUNCTION OF TIME.

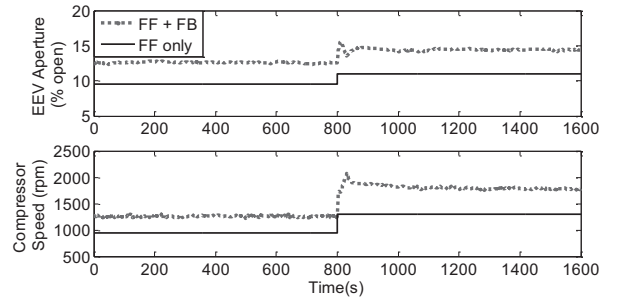


Figure 7. CONTROL INPUTS PLOTTED AS A FUNCTION OF TIME.

5 DISCUSSION

5.1 Comparison of Predicted and Experimental Results

As expected, there is some discrepancy between the predicted optimal operation and the actual system operation, as shown in Fig. 6, due to differences between the model and experimental system. The largest error is observed between T_1 and T_1^* in both Case 1 and Case 2. This is a result of the model uncertainty and assumptions associated with the estimation of $(UA)_{L,max}$ [9].

Given that Λ was chosen so that linear combinations of P_1 and P_2 were regulated using a feedback controller, it was ex-

pected that the optimal values of P_1 and P_2 would be achieved during operation of the experimental system. However, the effect of uncertainties associated with the empirical models for predicting K_v and η_{vol} , and consequently the need for linear feedback control, is evident in Fig. 7.

5.2 Comparison of Experimental Results Against Design Point Operation

Despite the potential for discrepancy, the experimental results illustrate that there is a significant benefit to the approach being advocated here. This is evident when we examine the results in the context of current industrial operation in which the operation set points are not optimized for varying cooling capacity but are instead chosen on, or near, the design point of the system albeit the system often operates at partial-load conditions.

In order to provide a reasonable condition for comparison, the experimental VCC system was operated as it might be if it were a commercial system - with a fixed speed chosen close to the maximum rated speed of the compressor, and a constant evaporator superheat of 10°C , typically regulated via a thermostatic expansion valve. This can be considered the most efficient the system would be operated in industry as it is near the design point of the system. Additional details are provided in Appendix B.

Table 4. COMPARISON OF OPTIMIZED OPERATION WITH DESIGN POINT OPERATION

Parameter	Units	Design Point	Case 1	Case 2
Capacity Achieved	kW	0.899	0.780	0.889
COP	—	1.84	2.42	1.97
η_{II}	%	7.93	9.01	8.33
T_H	$^\circ\text{C}$	23.4	23.3	23.5
T_L	$^\circ\text{C}$	11.1	12.7	11.5

For Case 1, in which the system was optimized to achieve 0.7kW of cooling capacity, the optimization results show a 31.5% increase in COP and a 13.6% increase in η_{II} over the system operating at its design point. This demonstrates that when operated off the design point, the system achieved a higher system efficiency with the use of optimized operation set points and flexible system control.

For Case 2, in which $C_{desired} = 0.85\text{kW}$, the system actually achieved 0.889kW of cooling capacity, nearly the same capacity that was achieved when the system was operated at its design point. Table 4 shows that when operated at optimized set points, a 7.07% increase in COP and a 5.04% increase in η_{II} , over design point operation, were achieved. Therefore, even at the design point operating capacity, the optimized set points in conjunction

with the feedforward plus feedback control architecture enabled more efficient operation of the system. However, as evidenced by the results in the two cases, the benefits were greater with greater deviation from the design point.

6 CONCLUSION

In this paper the authors considered the implementation of an exergy-based MDOF optimization and control methodology for steady-state operation of VCC systems. A dynamical analysis was conducted to characterize the control problem for the standard VCC, and a design variable, Λ , was introduced to provide flexibility in choosing the projection of the resulting 5 optimization variables onto the 4-dimensional control space. A model-based feedforward plus feedback control architecture was then proposed for operating a given system at its optimized operation set points. Experimental implementation of the optimization and control scheme demonstrated increases in system COP and second law efficiency, η_{II} , for two different cooling capacities. The results demonstrate that although VCC systems are designed well to operate at their design point, the proposed optimization framework can lead to improvements in energy efficiency while operating away from the design point. Future work will consider extending the optimization beyond steady-state operation.

Appendix A: System Identification

The open-loop data was detrended prior to identification; consequently all input and output variables are deviation variables with respect to the nominal operating condition of $u_1 = a_v = 11\%$ and $u_2 = \omega_k = 1200\text{ rpm}$. Additionally, the input and output data sets were scaled such that

$$\bar{u} = N_u \cdot u \quad (32)$$

$$\bar{y} = N_y^{-1} \cdot y \quad (33)$$

$$N_u = \begin{bmatrix} 1.87 & 0 \\ 0 & 0.0178 \end{bmatrix} \quad (34)$$

$$N_y = \begin{bmatrix} 18.7 & 0 \\ 0 & 9.93 \end{bmatrix} \quad (35)$$

where $\bar{u} \in \Re^{N \times 2}$ and $\bar{y} \in \Re^{N \times 2}$ represent the scaled input and output data sets, respectively, and N is the number of data points collected for the identification. The diagonal elements of N_u and N_y , respectively, are given as

$$N_u(i, i) = \sigma(u_i) \quad (36)$$

$$N_y(i, i) = \sigma(y_i) \quad (37)$$

for $u_i \in \Re^N$, $y_i \in \Re^N$, $i = 1, 2$, and where σ is the standard deviation. The scaled identified state space $[A, B, C, D]$ system model

is given in Eq. (38). Again, a bar denotes a scaled quantity.

$$\bar{A} = \begin{bmatrix} -0.00975 & 0.00266 \\ 0.0214 & -0.0240 \end{bmatrix}, \bar{B} = \begin{bmatrix} 0.000435 & -0.000200 \\ -0.00136 & 0.00103 \end{bmatrix}$$

$$\bar{C} = \begin{bmatrix} 37.1 & 3.63 \\ 17.2 & 28.7 \end{bmatrix}, \bar{D} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (38)$$

Appendix B: Baseline Operating Condition

A proportional-integral controller tuned for the experimental VCC system [21] was used to regulate evaporator superheat via an EEV (Fig. 8) to mimic superheat regulation via a thermostatic expansion valve, a mechanical control valve commonly used in industry. The compressor was operated at a fixed speed of 2000 rpm, and the evaporator and condenser fans were operated at 100% of their maximum power. The COP and second-law efficiency of the system operating at this condition are given in Tab. 4.

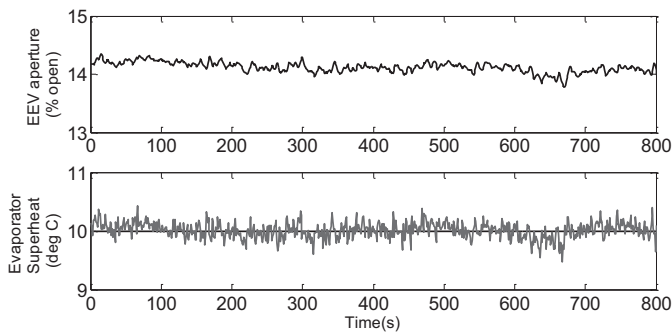


Figure 8. EEV CONTROL SIGNAL AND EVAPORATOR SUPERHEAT PLOTTED AS A FUNCTION OF TIME FOR $\omega_k=2000$ RPM.

ACKNOWLEDGMENTS

This research was supported in part by an award from the Department of Energy (DOE) Office of Science Graduate Fellowship Program (DOE SCGF).

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